

Problem Statement #40 — Student Startup Crowdfunding Platform

Retail, E-Commerce & Finance

This document contains the complete Lab 1 submission: the requirements table, actor list, use-case list, UML use-case diagram, include/extend relationship explanations, and the detailed use-case flow specification for the Student Startup Crowdfunding Platform.

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A. Requirements Table

Five Functional Requirements (FR-001–FR-005) and two Nonfunctional Requirements (NFR-001–NFR-002). FR-001 and NFR-001 (shaded) were provided in the assigned scenario and are preserved as given.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall collect backer pledges and disburse funds to the student venture in tranches only upon verification of predetermined project milestones.	High	Pass: Milestone evidence reviewed and tranche released. Fail: Campaign funds disbursed without milestone sign-off.	Protects backers by tying every fund release to verified progress — the core trust mechanism of the platform.
FR-002	Functional	The system shall allow a Student Founder to create and submit a crowdfunding campaign, including a funding target, milestone schedule, and campaign description, for platform approval.	High	Pass: A submitted campaign appears with status “Pending Review” and contains a funding target, description, and at least one milestone. Fail: A campaign is saved without a funding target or without any milestone defined.	Campaign creation is the entry point for every student venture; no funding activity can occur without it.
FR-003	Functional	The system shall allow a Campaign Backer to view the list of approved, currently active campaigns, including funding target, amount raised, and milestone status.	Medium	Pass: The “Browse Campaigns” view lists only campaigns with status “Approved” and shows amount raised versus target. Fail: A campaign with status “Pending Review” or “Rejected” appears in the public list.	Backers need accurate, trustworthy visibility into live campaigns before deciding to pledge.
FR-004	Functional	The system shall allow a Campaign Backer to submit a pledge toward an approved campaign and shall process the associated payment through the payment gateway before confirming the pledge.	High	Pass: A pledge is recorded and added to the campaign total only after the payment gateway returns an authorized transaction status. Fail: A pledge is recorded while the payment status is declined or still pending.	Ensures pledged funds are actually collected and tied to a valid, authorized transaction before being counted toward a campaign.
FR-005	Functional	The system shall allow the Faculty Committee to review submitted milestone evidence and record an approval or rejection decision for each milestone.	High	Pass: A milestone’s status changes to “Verified” or “Rejected” only after a Faculty Committee member records a decision, with reviewer ID and timestamp logged. Fail: A milestone status changes	Faculty verification is the checkpoint that gives backers confidence funds are released only for genuine progress,

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				without an associated committee decision record.	directly supporting FR-001.
NFR-001	Nonfunctional (Performance & Security)	The payment processing gateway must comply with PCI-DSS guidelines with tokenized payment authorizations.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Payment data is highly sensitive; regulatory compliance and tokenization are non-negotiable for handling real money.
NFR-002	Nonfunctional (Security & Privacy)	The system shall encrypt all backer personal information and payment-related metadata at rest and in transit, and shall restrict milestone evidence and financial data to authenticated users holding the appropriate role.	High	Pass: A security audit confirms personal/financial fields are stored encrypted (e.g., AES-256) and role-based access control blocks unauthorized read attempts. Fail: Any personal or financial field is found in plaintext, or a user accesses another actor's data without the correct role.	Protects student and backer privacy and maintains platform trust and compliance beyond the payment-specific scope of NFR-001.

B. Actor List

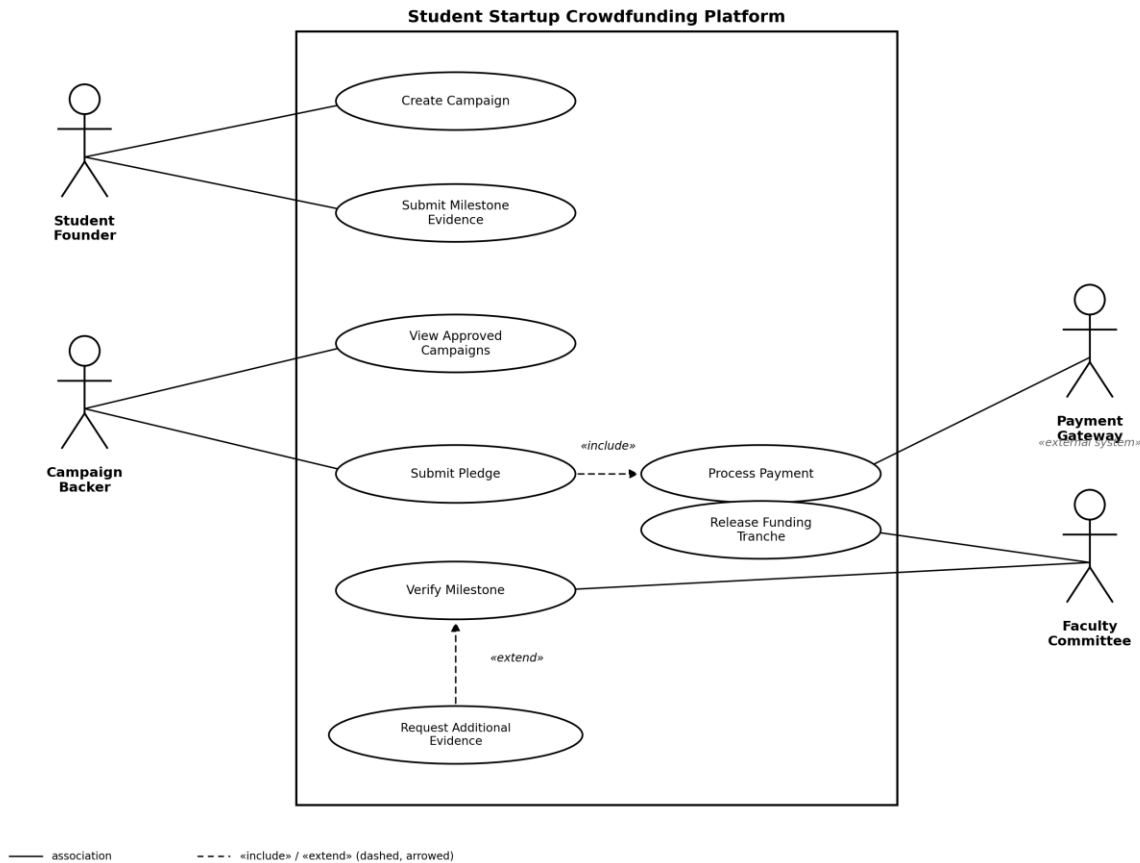
Actor	Why this actor interacts with the system
Student Founder	Creates crowdfunding campaigns and submits milestone evidence to request release of the next funding tranche.
Campaign Backer	Browses approved campaigns and submits pledges (payments) to fund student ventures they want to support.
Faculty Committee	Reviews and verifies milestone evidence submitted by Student Founders before any funding tranche is released.
Payment Gateway (external system)	An external payment-processing system that authorizes and settles pledge payments on behalf of the platform; it is outside the system boundary.

C. Use-Case List

Use Case ID	Use Case Name	Primary Actor	Purpose
UC-01	Create Campaign	Student Founder	Define and submit a new campaign (target, milestones, description) for platform approval.
UC-02	Submit Milestone Evidence	Student Founder	Upload proof that a project milestone has been completed, for Faculty Committee review.
UC-03	View Approved Campaigns	Campaign Backer	Browse the list of approved, currently active campaigns and their funding progress.
UC-04	Submit Pledge	Campaign Backer	Pledge funds toward an approved campaign; requires a successfully processed payment.
UC-05	Process Payment	Payment Gateway	Authorize and process the payment associated with a backer's pledge.
UC-06	Verify Milestone	Faculty Committee	Review submitted milestone evidence and record an approval or rejection decision.
UC-07	Release Funding Tranche	Faculty Committee	Release the funding tranche tied to a milestone once it has been verified.
UC-08	Request Additional Evidence	Faculty Committee	Ask the Student Founder for more proof when submitted evidence is judged insufficient (extends Verify Milestone).

D. UML Use-Case Diagram

System boundary: “Student Startup Crowdfunding Platform”. All four actors sit outside the boundary; all eight use cases sit inside it.



E. UML Relationship Explanation

«include» — Submit Pledge → Process Payment: every pledge submission must always invoke payment processing as a mandatory, reused part of its behaviour — a pledge is never completed without it, which is exactly what «include» models.

«extend» — Request Additional Evidence → Verify Milestone: requesting more evidence only happens conditionally, when the Faculty Committee judges the submitted evidence insufficient during a specific run of Verify Milestone, not on every verification — which is exactly what «extend» models.

F. Use-Case Flow Specification

Use Case ID	UC-06
Use Case Name	Verify Milestone
Primary Actor	Faculty Committee
Brief Description	Enables the Faculty Committee to review milestone evidence submitted by a Student Founder and record a verification decision that determines whether the corresponding funding tranche is released.

Preconditions

- The Student Founder has submitted milestone evidence for a specific campaign milestone (UC-02 – Submit Milestone Evidence).
- The Faculty Committee member is authenticated with committee-review privileges.
- The milestone status is “Submitted”, i.e. awaiting review.

Main Success Scenario

1. Faculty Committee opens the list of milestones awaiting review.
2. System displays the submitted milestone evidence and campaign details.
3. Faculty Committee reviews the evidence against the milestone’s predetermined criteria.
4. Faculty Committee approves the milestone.
5. System records the approval decision, reviewer ID, and timestamp.
6. System updates the milestone status to “Verified”.
7. System releases the corresponding funding tranche to the Student Founder (UC-07 – Release Funding Tranche).
8. System notifies the Student Founder that the milestone was verified and funds released.

Alternate Flow — A1: Milestone Evidence Insufficient

1. Faculty Committee determines the submitted evidence does not satisfy the milestone criteria.
2. Faculty Committee selects “Request Additional Evidence” (UC-08).
3. System records the insufficient/rejected status together with the committee’s comments.
4. System notifies the Student Founder that additional evidence is required.
5. Student Founder submits revised evidence (UC-02); verification resumes at step 2 of the Main Success Scenario.

Postconditions

- Success: Milestone status is “Verified”, the funding tranche is released, and the Student Founder is notified — consistent with FR-001, since funds are never released without milestone verification.
- Alternate: Milestone status is “Evidence Requested”, no funds are released, and the Student Founder is notified to resubmit evidence.

G. Final Submission Checklist

- Exactly 5 Functional Requirements (FR-001–FR-005) and 2 Nonfunctional Requirements (NFR-001–NFR-002).
- FR-001 and NFR-001 preserved exactly as given in the assigned scenario.
- Every requirement has an ID, type, testable “shall” description, priority, measurable pass/fail acceptance criteria, and rationale.
- At least 3 actors identified (4 provided), each external to the system and each with a clear reason for interacting with the system.
- At least 5 use cases identified (8 provided), collectively covering every functional requirement.
- UML use-case diagram includes a labelled system boundary, all actors outside it, all use cases inside it, correct actor–use-case associations, at least one «include» relationship, and at least one «extend» relationship, all in standard UML notation.
- Include and extend relationship directions verified as UML-correct and briefly justified.
- One-page Use-Case Flow Specification provided for Verify Milestone, with preconditions, a numbered main success scenario, one alternate flow, and postconditions.
- The flow is consistent with FR-001: funds are never released without milestone verification.
- No placeholders remain; the scenario (Student Startup Crowdfunding Platform) was preserved throughout and not replaced with a generic example.

Use-Case Flow Specification — Student Startup Crowdfunding Platform

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